2201651-Experiment 10 ANOVA

1. Explain input and output characteristics for f_oneway function from scipy.stats library:
The f_oneway function in the scipy.stats library performs a one-way ANOVA (Analysis of Variance) test.

- Input: It accepts two or more arrays (or lists) of numerical data, each representing a sample from a different group.
  Example: f_oneway(group1, group2, group3)
- Output: It returns a statistic (F-value) and a p-value.
- F-value measures the ratio of variance between the groups to the variance within the groups.
- p-value indicates the significance of the results; a smaller value (typically < 0.05) suggests a statistically significant difference among the group means.

2. Explain results of sm.stats.anova_lm function:
The sm.stats.anova_lm() function from the statsmodels library is used to perform ANOVA on a linear model.

- It takes a fitted OLS model as input (e.g., ols_model = sm.OLS(y, X).fit()), and then anova_lm(ols_model) is called.
- Output: A table with columns like:
- df: degrees of freedom for each source
- sum_sq: sum of squares
- mean_sq: mean square (sum_sq / df)
- F: F-statistic
- PR(>F): p-value for the F-statistic
  The table helps determine whether any independent variables significantly affect the dependent variable.

3. What role ANOVA plays in data analytics?
ANOVA is a statistical method used to compare the means of multiple groups and determine if at least one group mean is significantly different.
In data analytics, ANOVA helps:

- Identify key variables or factors influencing outcomes
- Support decision-making with evidence
- Reduce noise by focusing on statistically significant variables
- Test hypotheses in A/B testing or experimental designs

4. When to use one-way ANOVA? Explain with real world case:
Use one-way ANOVA when you want to compare the means of three or more independent groups based on one categorical independent variable.
Real-world case: A teacher wants to compare the average exam scores of students from three different teaching methods (Traditional, Online, Hybrid). One-way ANOVA can test if the teaching method significantly affects student performance.

5. When to use two-way ANOVA? Explain with real world case:
Use two-way ANOVA when there are two independent categorical variables and you're analyzing their effect on one continuous dependent variable. It also helps identify interaction effects.
Real-world case: A company tests the productivity of employees based on work shift (day/night) and training program (A/B). Two-way ANOVA can analyze the effect of both factors and whether their combination impacts productivity.